

that help engineers design, test and deploy embedded code," Tony says.

Whereas Choudhury feels that the industrial automation industry is characterised by smaller volumes and a huge variety of applications servicing diverse needs. "Automation companies customise products for specific applications and end-user requirements. Hence innovation comes more from targeted applications and less so from the application of the hottest and the newest technologies," he adds.

"The one exception to that is on the sensor and the human-machine interface (HMI) side, where significant innovations and developments have resulted. Wide-ranging sensors for measurements as well as touch-screen interfaces are bringing growth to the technology providers," he says, adding, "Another unique characteristic of such automation is the decades of reliability and shelf life that the equipment and the resultant semiconductor parts have to endure. Automating a process or the factories is very expensive, and, once done, it stays for several years. Hence a highly configurable and flexible system is always desired besides the reliability."

Wadhwa opines that the automation products and solutions are oriented along three categories—Continuous process control applications (as in chemical, food and beverage, oil and gas, life sciences, traditional power generation, pulp and paper, wastewater, etc), job work and batch production applications (as in automotive, fabricated metals, plastics, packaging, etc) and electrical energy management application (transmission, distribution, etc).

Informing about the latest trends in the different segments of automation products, he thinks in the sensors and transmitters segment, there is a trend towards digital technologies with smart diagnostics and wireless connectivity; in the control valves and actuators, there is a trend towards higher precision electro-pneumatics and electrohydraulics; in AC/DC drive, the trend is towards digital ar-

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chitecture; in motion control systems, the trend is towards compactness, open standards and multiple-axis capability; in recorders, the trend is towards software-based and real-time analytics; in PLCs, the trend is towards smaller form factor, open connectivity and in-built safety; in DCS/SCADA, the trend is towards KPI dashboards, web-reporting features; and in MMI/HMI, there is an inclination towards colour touch screens.

In a broad framework, the new technologies in the field of industrial automation are as follows:

Industrial wireless. The application for wireless technology is very broad and it will bring the ability to do things that were not even thought of before. However, the most important thing for wireless sensor networks is the reliability, which can be ensured by combining meshing with spread-spectrum technology. Other factors, such as security, latency, throughput and power also contribute to the increasing use of industrial wireless technology.

Embedded intelligence. Embedded systems have a major role in the development of the Internet and wireless communication channels. M2M and pervasive computing are actually embedded intelligence and connectivity and they provide valuable information about assets, liabilities and costs to managers and help in the decision-making process, which can immensely help in growth of productivity and efficiencies.

Web services. Web services involve application-to-application communication and write to clients in any language that supports HTTP and SOAP. Usually, any sort of user interface is

not built-in in web services, and it is up to the client to process input and display output. It helps in collaboration amongst work groups and staff working in different plants.

Real-time applications. Real-time enterprise acts on events as they happen, and it is all about getting information in and out quickly, making quick and effective decisions and monitoring the business as it happens.

Robots. Robots with integrated vision and touch are very useful in the automation process. They have the capability to increase the speed and efficiency manifold. The latest robots are very efficient and accurate, and they can replace manual operations very easily.

Future outlook

Despite the not-so-upbeat trend worldwide, India has a lot of potential to grow in the automation sector. Indian machine builders are not only looking up to the domestic market but are also vying for export sales. However, they face tough competition from global leaders. So they need to enhance their production speed and quality besides reliability and safety as these factors will majorly contribute to the growth of the automation industry.

In the coming years, high-speed communication bus—EtherNet/IP, EtherCAT and Profinet—will be ruling their field, and remote communication and wireless communication will also play an important role. The sectors like packaging, textile, printing, pharma and food are expected to drive the growth.

However, to become a leader, the Indian automation industry will have to focus on price and technology. Though we are competitive in software development and services, we are far behind in becoming a technology leader, as we lack on the R&D and hardware points. On the price and skills front, we are good and we can be a leader in this aspect, and can also provide engineering and after-sales service. ●

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